respectively. If economy needs 100 tonnes of coal and 50 tonnes of steel, calculate the gross output of the two commodities and the total labour days required.

Solution

Here the technology matrix is given under

	Steel	Coal	Final demand
Steel	0.4	0.1	50
Coal	0.7	0.6	100
Labour days	5	2	-

The technology matrix is $B = \begin{bmatrix} 0.4 & 0.1 \\ 0.7 & 0.6 \end{bmatrix}$

$$I - B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0.4 & 0.1 \\ 0.7 & 0.6 \end{bmatrix}$$

$$= \begin{bmatrix} 0.6 & -0.1 \\ -0.7 & 0.4 \end{bmatrix}$$

$$|I - B| = \begin{vmatrix} 0.6 & -0.1 \\ -0.7 & 0.4 \end{vmatrix}$$

$$= (0.6)(0.4) - (-0.7)(-0.1)$$

$$= 0.24 - 0.07 = 0.17$$

Since the diagonal elements of $I - B$ are positive and value of $|I - B|$ is positive, the system is viable.

$$\text{adj } (I-B) = \begin{bmatrix} 0.4 & 0.1 \\ 0.7 & 0.6 \end{bmatrix}$$

$$(I-B)^{-1} = \frac{1}{|I - B|} adj(I - B)$$

$$= \frac{1}{0.17} \begin{bmatrix} 0.4 & 0.1 \\ 0.7 & 0.6 \end{bmatrix}$$

$$X = (I-B)^{-1}\text{D, where D} = \begin{bmatrix} 50 \\ 100 \end{bmatrix}$$

$$= \frac{1}{0.17} \begin{bmatrix} 0.4 & 0.1 \\ 0.7 & 0.6 \end{bmatrix} \begin{bmatrix} 50 \\ 100 \end{bmatrix}$$

$$= \frac{1}{0.17} \begin{bmatrix} 30 \\ 95 \end{bmatrix}$$

$$= \begin{bmatrix} 176.5 \\ 558.8 \end{bmatrix}$$

Steel output = 176.5 tonnes

Coal output = 558.8 tonnes

Total labour days required

= 5(steel output) + 2(coal output)

= 5(176.5) + 2(558.8)

= 882.5 + 1117.6 = 2000.1

$\simeq$ 2000 labour days.

Exercise 1.4

1. The technology matrix of an economic system of two industries is $\begin{bmatrix} 0.50 & 0.30 \\ 0.41 & 0.33 \end{bmatrix}$. Test whether the system is viable as per Hawkins Simon conditions.
2. The technology matrix of an economic system of two industries is $\begin{bmatrix} 0.6 & 0.9 \\ 0.20 & 0.80 \end{bmatrix}$. Test whether the system is viable as per Hawkins-Simon conditions.
3. The technology matrix of an economic system of two industries is $\begin{bmatrix} 0.50 & 0.25 \\ 0.40 & 0.67 \end{bmatrix}$. Test whether the system is viable as per Hawkins-Simon conditions.
4. Two commodities A and B are produced such that 0.4 tonne of A and 0.7 tonne of B are required to produce a tonne of A. Similarly 0.1 tonne of A and 0.7 tonne of B are needed to produce a tonne of B. Write down the technology matrix. If 68 tonnes of A and 10.2 tonnes of B are required, find the gross production of both of them.

01_11th_BM-STAT_Ch-1-EM.indd 19 21-04-2020 12:20:11 PM